

Numerical Solution to the Perturbation Equations

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The purpose of writing this document is to stepwisely derive the numerical solution of the set of perturbation equations of the TLN-TDN project.

1 Solution to the TOV Equation

TOV equation is a set of equation that describes the internal structure of the star, given a equation of state (EOS), which is a function that describes the relation between stellar pressure and density.

The TOV equation has the form:

$$\frac{dp}{dr} = -\frac{(p + \rho)(m + 4\pi r^3 p)}{r(r - 2m)} \quad (1.1)$$

Given a background metric:

$$ds^2 = -e^\nu dt^2 + e^\lambda dr^2 + r^2 d\Omega^2,$$

one is also able to obtain the relation:

$$\frac{d\nu}{dr} = 2\frac{m + 4\pi r^3 p}{r(r - 2m)} \quad (1.2)$$

Also note that we have the conservation equation:

$$\frac{dm}{dr} = 4\pi r^2 \rho \quad (1.3)$$

With this set of equation, we are able to solve the background stellar pressure and mass functions. This is a set of first-order ODEs, however, their initial condition are not independent.

We assume functional forms of the TOV equations to be polynomials, with max order 3 (by omitting the higher-order contribution, and the r contribution to the mass function is in the third order):

$$\nu(r) = a_0 + a_1 r + a_2 r^2 + a_3 r^3 \quad (1.4)$$

$$m(r) = b_1 r + b_2 r^2 + b_3 r^3 \quad (1.5)$$

$$p(r) = c_0 + c_1 r + c_2 r^2 + c_3 r^3 \quad (1.6)$$

We have set $b_0 = 0$ as the initial condition for the mass function.

Substitute the polynomials back to the set of equations, and expand them to the second order of r . The mass conservation yields:

$$b_1 + r \left(2b_2 + 3b_3 r - 4\pi r \left(\frac{c_0}{K} \right)^{n/(n+1)} \right) = 0, \quad (1.7)$$

Which indicates:

$$b_1 = b_2 = 0, b_3 = \frac{4\pi}{3} \left(\frac{c_0}{K} \right)^{n/(n+1)} \quad (1.8)$$

Solve for c_0 we obtain:

$$c_0 = K \left(\frac{3b_3}{4\pi} \right)^{1+\frac{1}{n}} \quad (1.9)$$

Given that $b_1 = b_2 = 0$, the $d\nu/dr$ relation gives:

$$a_1 + r(2a_2 - 2b_3 - 8\pi c_0) + r^2(3a_3 - 8\pi c_1) = 0 \quad (1.10)$$

We have:

$$a_1 = 0, \quad c_0 = \frac{a_2 - b_2}{4\pi}, \quad c_1 = \frac{3}{8\pi} a_3 \quad (1.11)$$

With $a_1 = 0$, we obtain:

$$\begin{aligned} c_1 + r \left(b_3 \left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} + b_3 c_0 + 4\pi c_0^2 + 4\pi c_0 \left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} + 2c_2 \right) \\ + r^2 \left(\frac{b_3 c_1 n \left(\frac{c_0}{K} \right)^{\frac{n}{n+1}}}{c_0(n+1)} + b_3 c_1 + \frac{4\pi c_1 (2n+1) \left(\frac{c_0}{K} \right)^{\frac{n}{n+1}}}{n+1} + 8\pi c_0 c_1 + 3c_3 \right) = 0 \end{aligned} \quad (1.12)$$

We observe that $c_1 = 0$ and therefore $a_3 = 0$. We substitute this result back into Equation 1.12 and obtain:

$$r \left(b_3 \left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} + b_3 c_0 + 4\pi c_0^2 + 4\pi c_0 \left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} + 2c_2 \right) + 3c_3 r^2 = 0 \quad (1.13)$$

We obtain that $c_3 = 0$ and a rather complicated relation between c_0, c_2, b_3 , but with Equation 1.8:

$$c_2 = -\frac{1}{32} \pi c_0 (9\pi c_0 K + 64) \left(\left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} + c_0 \right) \quad (1.14)$$

To summarize:

$$b_1 = b_2 = c_1 = 0; \quad c_0 = K \left(\frac{3b_3}{4\pi} \right)^{1+\frac{1}{n}} = \frac{a_2 - b_2}{4\pi} \quad (1.15)$$

$$c_2 = -\frac{1}{32} \pi c_0 (9\pi c_0 K + 64) \left(\left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} + c_0 \right) \quad (1.16)$$

With the hidden relation:

$$\frac{dp}{dr} = -\frac{1}{2(p+\rho)} \frac{d\nu}{dr}, \quad (1.17)$$

we are also able to obtain:

$$a_1 = 0, \quad a_2 + 2c_2 \left(\left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} + c_0 \right) = 0, \quad a_3 + 2c_3 \left(c_0 + \left(\frac{c_0}{K} \right)^{\frac{n}{n+1}} \right) = 0 \quad (1.18)$$

We are then able to reduce the ansatz to the form:

$$\nu(r) = a_0 + a_2 r^2 + a_3 r^3 \quad (1.19)$$

$$m(r) = \frac{4\pi}{3} \left(\frac{c_0}{K} \right)^{n/(n+1)} r^3 \quad (1.20)$$

$$p(r) = c_0 + c_2 r^2 + c_3 r^3 \quad (1.21)$$

I think it's not really possible to obtain a relation between a_0 and c_0 from this set of equations, as there is no ν term and a_0 is always eliminated in the ν' term.

To match this result, we compare to the asymptotic behaviors in [1302.1918](#).

$$\rho(R) = \rho_c + \rho_2 R^2 + \mathcal{O}(R^3) \quad (1.22)$$

$$p(R) = p_c + \frac{2\pi}{3} (\rho_c + p_c) (\rho_c + 3p_c) R^2 + \mathcal{O}(R^3) \quad (1.23)$$

$$M(R) = \frac{4\pi}{3} R^3 + \frac{4\pi}{5} \rho_2 R^5 + \mathcal{O}(R^6) \quad (1.24)$$

$$\nu(R) = \nu_c + \frac{4\pi}{3} (\rho_c + 3p_c) R^2 + \mathcal{O}(R^3) \quad (1.25)$$

We recognize that with the polytropic EOS, $\frac{4\pi}{3} (p_0/K)^{n/(n+1)}$ is the central density, and the terms of my calculation in general matches the result in the paper.

1.1 A dimensionless equation?

Since there aren't fixed relation between a_0 and c_0 , but we still want initial condition for ν . We therefore consider the stellar surface, where the exterior solution satisfies the birkhoff's theorem, meaning the external metric satisfies the Schwarzschild solution.

We introduce new variables:

$$\bar{r} := \frac{r}{R}, \quad \bar{m} := \frac{m}{M}, \quad \bar{P} = \frac{p}{p_c} \quad (1.26)$$

With a polytropic EOS:

$$p = K\rho^{1+1/n} \quad \Rightarrow \quad \rho = \left(\frac{p}{K}\right)^{n/(n+1)} \quad (1.27)$$

2 Solution to the set of perturbation equations

The set of perturbation so solve attains the form:

$$H_1' = \frac{e^\lambda}{r} H + \left[4\pi r (\rho - p) e^\lambda - \frac{2me^\lambda}{r^2} - \frac{\ell+1}{r} \right] H_1 + \frac{e^\lambda}{r} K - 16\pi \frac{\rho+p}{r} e^\lambda V - 32\pi i\omega\eta \frac{e^{\lambda-\nu/2}}{r} V, \quad (2.1)$$

$$K' = \frac{H}{r} + \frac{\ell(\ell+1)}{2r} H_1 + \left(\frac{\nu'}{2} - \frac{\ell+1}{r} \right) K - 8\pi \frac{\rho+p}{r} e^{\lambda/2} W - 32\pi i\omega\eta \frac{e^{-\nu/2}}{r} V, \quad (2.2)$$

$$W' = \frac{r}{2} e^{\lambda/2} H + r e^{\lambda/2} K + \frac{r e^{(\lambda-\nu)/2}}{\rho+p} c_s^{-2} X - \frac{\ell+1}{r} W - \frac{\ell(\ell+1)}{r} e^{\lambda/2} V - 16\pi i\omega\eta r e^{(\lambda-\nu)/2} V, \quad (2.3)$$

$$\begin{aligned} X' = & \frac{\rho+p}{2} e^{\nu/2} \left(\frac{1}{r} - \frac{\nu'}{2} \right) H + \frac{\rho+p}{2} e^{\nu/2} \left[\omega^2 r e^{-\nu} + \frac{\ell(\ell+1)}{2r} \right] H_1 + \frac{\rho+p}{2} e^{\nu/2} \left(\frac{3\nu'}{2} - \frac{1}{r} \right) K \\ & - \frac{\ell}{r} X - \frac{\rho+p}{r} e^{(\lambda+\nu)/2} \left[\omega^2 e^{-\nu} + \frac{e^{-\lambda}}{4r^2} \left\{ e^{2\lambda} (1+8\pi r^2 p)^2 + 2e^\lambda (3+8\pi r^2 p) - 7 \right\} \right] W + \frac{\ell(\ell+1)}{r^2} e^{\nu/2} p' V \\ & + \alpha_\eta \eta + \alpha_{\eta'} \eta' + \alpha_\zeta \zeta + \alpha_{\zeta'} \zeta' \end{aligned} \quad (2.4)$$

Where we have above solved the numerical profile for ρ, p, ν and λ .